1 Planning (15 marks)

Defining the problem (3 marks)

P *v* is the independent variable or vary *v.* [1]

P *E* is the dependent variable or measure *E.* [1]

P Keep the number of turns on the coil constant. [1]

Methods of data collection (5 marks)

M1 Labelled diagram showing magnet falling vertically through coil. [1]

M2 Voltmeter or c.r.o. connected to the coil. Allow voltage sensor connected to datalogger. [1]

M3 Method to change speed e.g. change height. [1]

M4 Measurements to determine *v*. Use metre rule to measure distance magnet falls to the bottom of the coil or metre rule/ruler to measure length of coil or ruler to measure length of the magnet. [Allow timing instrument to measure the time of the fall from the start to the bottom of the coil.] [1]

M5 Method of determining *v* corresponding to appropriate distance e.g. $v = \sqrt{2gh}$ or $v=2h/t$ (for height method) or $v = L/t$ for length of magnet or coil and by stopwatch, timer or lightgate(s) connected to datalogger. [Allow $v = gt$ for timing fall to bottom of coil.] [1]

Method of analysis (2 marks)

A Plot a graph of *E* against *v*. [Allow lg *E* against lg *v*] [1]

A Relationship valid if straight line through origin. [1]
[If lg-lg then straight line with gradient = (+)1 (ignore reference to *y*-intercept)]

Safety considerations (1 mark)

S Keep away from falling magnet/use sand tray/cushion to catch magnet. [1]

Additional detail (4 marks)

D1/2/3/4 Relevant points might include [4]

Use coil with large number of turns/drop magnet from large heights/strong magnet

1 Detailed use of datalogger/storage oscilloscope to determine maximum *E*; allow video camera including slow motion play back
2 Use same magnet or magnet of same strength.
3 Use of short magnet so that *v* is (nearly) constant
4 Use short/thin coil so that *v* is (nearly) constant
5 Use a non-metallic vertical guide/tube
6 Method to support vertical coil or guide/tube
7 Repeat experiment for each *v* and average

Do not allow vague computer methods.

[Total: 15]

© Cambridge International Examinations 2012

https://xtremepape.rs/